

# Rohan Ferriols Alibutud

PhD Candidate, Bioinformatics  
Temple University

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## EDUCATION

### **PhD Candidate (*in progress*)** **2023 - 2026**

Bioinformatics program  
Temple University Department of Biology  
*Advisor:* Sudhir Kumar

### **Bachelor of Science *cum laude*** **2015 - 2020**

Double major, Genetics and Evolutionary Anthropology  
Rutgers, The State University of New Jersey  
*Thesis:* Prioritization of autism candidate genes from whole-genome sequences of affected families.  
*Thesis advisors:* Jinchuan Xing, Erin Vogel

### **Field School Graduate** **2018**

Turkana Basin Institute Origins Field School

### **Emergency Medical Technician (EMT) Certification** **2015**

Jersey Shore University Medical Center

### **High School Graduate** **2015**

Academy of Allied Health and Science

## RESEARCH

### **Doctoral Student** **2023 - 2026**

Kumar Lab  
Institute for Genomic and Evolutionary Medicine  
Temple University Department of Biology

### **Research Assistant** **2020 - 2023**

Xing Lab of Genomics  
Rutgers University Department of Genetics

Analyzed structural variants in autism probands using computational methods. Formulated a prioritization pipeline for copy number variants using Python, SQL, and R. Resulted in two publications, including one paper as a first author (see *Publications*).

## Undergraduate Student

2019 - 2020

Xing Lab of Genomics

Rutgers University Department of Genetics

Used bioinformatics tools and applied custom scripts written in Python and R. Prioritized target autism genes for induced pluripotent stem cell research conducted by collaborators. Resulted in an undergraduate honors thesis (see *Publications*).

## Database Assistant

2019

Vogel Lab

Rutgers University Department of Anthropology

Processed foraging data from Kalimantan field site using Excel, Access, and Visual Basic.

## MEMBERSHIPS IN SCIENTIFIC AND PROFESSIONAL ORGANIZATIONS

### Student Member

2026

Society for Molecular Biology and Evolution

### Undergraduate Affiliate

2020

Center for Human Evolutionary Studies

Rutgers University

## PUBLICATIONS

### In review or submitted

1. Kumar, S., **Alibutud, R.**, Allard, J., Mowlaei, M.E., and Shi, X., 2026. "Transforming molecular evolutionary analyses with protein foundation models". *Nature Methods* [submitted].
2. **Alibutud, R.** & Kumar, S. Multiple versus pairwise sequence alignments for protein phylogenetics using foundation models. *bioRxiv* 2026.05.26.727927 (2026) doi:10.64898/2026.05.26.727927.

### Published

3. Prem, S., Dev, B., Peng, C., Mehta, M., **Alibutud, R.**, Connacher, R.J., St Thomas, M., Zhou, X., Matteson, P., Xing, J. and Millonig, J.H., 2024. Dysregulation of mTOR signaling mediates common neurite and migration defects in both idiopathic and 16p11. 2 deletion autism neural precursor cells. *Elife*, 13, p.e82809.
4. **Alibutud, R.**, Hansali, S., Cao, X., Zhou, A., Mahaganapathy, V., Azaro, M., Gwin, C., Wilson, S., Buyske, S., Bartlett, C., Flax, J., Brzustowicz, L., Xing, J., 2023. Structural variations contribute to the genetic etiology of autism spectrum disorder and language impairments. *Int. J. Mol. Sci.* 24, 13248.
5. **Alibutud, R.** Conference paper. International Conference on Intelligent Biology in Medicine 2022. "Functional Impact of Copy Number Variation in Autism Probands". Philadelphia, PA.
6. **Alibutud, R.** "Prioritization of autism candidate genes from whole-genome sequences of affected families". Honors thesis, Rutgers University Department of Anthropology. Thesis advisors: Jinchuan Xing, Erin Vogel

### CONFERENCE PRESENTATIONS

- |                                                                                                                                      |             |
|--------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Alibutud, R.</b> "Molecular phylogenetics directly from deep learning protein foundation models". SMBE 2026, Copenhagen, Denmark. | <b>2026</b> |
| <b>Alibutud, R.</b> "Functional impact of copy number variants in autism probands". ICIBM 2022, Philadelphia, PA.                    | <b>2022</b> |

### TEACHING EXPERIENCE

#### Teaching Assistant

- |                                                     |                    |
|-----------------------------------------------------|--------------------|
| Bruce Weir Summer Institute in Statistical Genetics | <b>2025 - 2026</b> |
| BIOL-3112 Genomic Evolutionary Medicine             | <b>2024 - 2026</b> |
| BIOL-1001 Biology Lab for Non-Majors                | <b>2023</b>        |

### HONORS AND AWARDS

- |                                                                                                                    |             |
|--------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Outstanding Teaching Assistant by a Graduate Student</b><br>Temple University College of Science and Technology | <b>2024</b> |
| <b>Google Computing Resources Grant</b>                                                                            | <b>2024</b> |

|                                                           |                    |
|-----------------------------------------------------------|--------------------|
| <b>ICIBM 2022 NSF Travel Award</b>                        | <b>2022</b>        |
| <b>Honors Symposium in Anthropology First Place Award</b> | <b>2020</b>        |
| <b>Henry Rutgers Scholar</b>                              | <b>2020</b>        |
| <b>Paul Robeson Scholar</b>                               | <b>2020</b>        |
| <b>Dean's List</b>                                        | <b>2017 - 2020</b> |

## **EXTRACURRICULAR**

### **President, Rutgers Archaeological Society** **2018 - 2019**

Coordinated field trips to museums and archaeological sites and organized fossil hunting expeditions. Hosted guest lecturers on ongoing research in anthropology.

### **Media Coordinator, NJPIRG** **2015**

Liaised with local reporters on a solar energy campaign and organized tabling sessions and petitions.